Rishabh Jain

Senior Software Engineer | Distributed Systems, Applied AI and Data Science

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PROFESSIONAL EXPERIENCE

Microsoft

2024 to present

Senior Software Engineer | Azure SQL Database

- Improved resource allocation through differential overbooking and CPU-rebalancing changes, supported by metrics and dashboards for resource imbalance and rebalancing effectiveness.
- Implemented NUMA-aware CPU allocation for containerised workloads, integrating hardware topology into resource-governance workflows.
- Designed failover diagnostics and automated alerting across engineering teams. Investigated production incidents using telemetry and memory dumps, and fixed crash and exception-handling defects.
- Built a beta AI-assisted tool for incident investigation, operational reporting and deployment correlation, grounded in existing diagnostic queries and investigation guidance.
- Built a local AI-assisted code-analysis tool and delivered fixes for latent boundary, arithmetic and resource-allocation defects.

Technologies: C++, Azure SQL, Kusto (KQL) and WinDbg.

Google

2021 to 2024

Software Engineer III (L4) | Encrypted Editors: Docs, Sheets and Slides

- Led a seven-engineer team building encrypted import and export infrastructure and improving document fidelity.
- Designed and implemented Excel (.xlsx and .xls) imports for encrypted Sheets and Word (.docx) imports for encrypted Docs using Web Workers.
- Implemented Excel exports for encrypted documents through Google Takeout. Profiled and improved import and export performance against server-side processing.
- Conducted 50+ technical interviews, managed an intern and led Women Engineers workshops.

Technologies: Java, Web Workers and Spanner.

Microsoft

2018 to 2021

Software Engineer II | Microsoft Teams

- Contributed to notification services processing 2.1 billion events daily and scaled Redis caching without downtime.
- Reduced partner network calls by 60%. Extended cache lifetimes with more precise invalidation, consolidated APIs and reordered requests to avoid unnecessary downstream calls.
- Added customer-level encryption in Azure Storage Queues and improved resilience through stale caching and data-driven retry policies.
- Supported business continuity and disaster recovery testing and monitoring dashboards. Added reaction, shared and priority notifications, and analysed notification-email click-through rates.

Technologies: .NET Framework, .NET Core, Azure Storage and Redis.

Kuliza | Software Developer Intern

2018

Developed asynchronous caching, role-based access controls, and lead generation and aggregation for HDFC's brand website and CMS administration.

HackerRank | Software Engineering Intern

June to July 2017

Created SQL, algorithm and regular-expression challenges for hiring assessments.

TECHNICAL SKILLS

Languages: C++, Java, C#, Python, SQL and JavaScript.

Backend and cloud: .NET Framework, .NET Core, Guice, Azure Cloud Services and Service Fabric.

Data and messaging: Azure SQL, Spanner, Cosmos DB, Redis, Azure Tables, Blob Storage and Storage Queues.

Analysis and ML: Kusto (KQL), pandas, NumPy, GeoPandas, LightGBM, Keras, TensorFlow, Chroma and Ragas.

Development and diagnostics: Git, Docker, WinDbg, Jupyter, Matplotlib and Seaborn.

PROJECT WORK

Legal Document RAG (LegalBench-RAG 2024)

2026

Python, Chroma, OpenAI and Ragas | Report | Repository

- Built contract-scoped retrieval over 644 contracts and 63,914 chunks, with cited answers and abstention. Retrieved reference evidence for nine of ten sampled benchmark questions (Hit@5: 0.90).

How Delhi Buses Move

2026

Python, GTFS-Realtime, pandas and LightGBM | Report | Repository

- Processed six hours of GPS from 6,065 buses for trajectory analysis and arrival modelling. Reduced arrival-time mean absolute error by 14% against a historical-median baseline in a time-matched evaluation.

Demand-Supply Workforce Heatmaps

2019; reviewed in 2026

Android, Django and SQLite; separate LSTM experiment | Report | Repository

- Built a team prototype comparing synthetic demand and resources across ten Bengaluru zones. Placed third nationally at Zomato Hackanoodle and presented to Deepinder Goyal. Added regression tests and an interactive report in 2026.

NYC Yellow Taxi Analysis

2026

Python, pandas and GeoPandas | Report | Repository

- Analysed a 0.75% sample from 38.3 million 2023 trip records to examine demand, geography and fares. Published an interactive report with data-quality checks and limitations on recorded-tip analysis.

EDUCATION

MS in Machine Learning and Artificial Intelligence

2025 to present

Liverpool John Moores University, via UpGrad. In progress.

Executive Diploma in Machine Learning and AI

2025 to present

IIIT Bangalore, via UpGrad. In progress.

B.Tech in Computer Science and Engineering

2014 to 2018

Jaypee Institute of Information Technology, Noida. Honours; CGPA: 8.9.

MENTORING

- Mentored 15+ learners over 100+ hours through Scaler Academy, 10xAcademy and other programmes, covering system design, backend development and interview preparation.
- Conducted mock interviews and supported mentees who secured roles at organisations including Amazon, Google and Microsoft.

AWARDS AND RECOGNITION

- Received the Google Cloud Tech Impact Award in 2022 for client-side encryption work.
- Received three Google Grungy Awards for addressing technical debt, infrastructure issues and reliability.
- Represented JIIT at ICPC Amritapuri in 2017 and achieved 97th All India Rank.